ModelIt!

Student Activity Pack

# Can We Actually Feed 10 Billion People?

Modeling Global Agricultural Systems and Food Security

Grade 11L1 | NGSS HS-LS2-1, HS-ESS3-3 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# Table of Contents

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1 Pre-Assessment ..... 3

2 Scoring Rubric ..... 4

3 Component Analysis ..... 5

4 Model Recording Page ..... 6

5 Simulation Observations ..... 7

6 Research & Extend ..... 8

7 Reflection Questions ..... 9

8 STEM Challenge ..... 10

# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where does your food come from? Trace one item from your last meal back to the farm — how many steps and resources were involved?**

2. **Do you think the world currently produces enough food to feed everyone? If so, why are hundreds of millions of people still hungry?**

3. **What do you think would happen to the environment if we doubled the amount of farmland in the world?**

4. **Draw a diagram showing what you think are the most important inputs needed to grow food and the most important outputs and waste products.**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Arable Land Area | Fertilizer Input | Water Availability | Crop Yield per Hectare | Soil Health Index | Total Food Production

| EXTERNAL | INTERNAL |
|---|---|
| | |
| | |
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Green Revolution Intensification

Fertilizer: Maximum | Water: Maximum | Land: Current

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Land Expansion

Arable land: +30% from forest conversion | Current inputs

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Sustainable Intensification

Fertilizer: Moderate | Soil conservation: Active | Land: Current

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Climate Change Impact** — The effect of rising temperatures, shifting rainfall patterns, and extreme weather events on crop yields — models predict yield declines of 2-6% per decade in tropical regions even as demand increases

**Diet Composition** — The global balance between plant-based and animal-based diets — producing one calorie of beef requires approximately 25 calories of grain feed, making diet shifts a powerful lever for food security

**Food Distribution Equity** — The fairness of food access across regions and income levels — the world produces enough calories for 10+ billion people, but distribution failures leave 735 million chronically hungry while wealthy nations waste 30-40% of their food

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**The world will have 10 billion mouths to feed by 2058 — but we're already using half the planet's habitable land for farming. Can we grow enough food without destroying the ecosystems we depend on?** Use evidence from your simulation to support your answers.

1. **How does your model reveal the tension between maximizing short-term crop yield and maintaining long-term soil health?**

2. **What surprised you most about the relationship between fertilizer use and soil degradation over time?**

3. **How does the model help explain why the world produces enough food for everyone but hundreds of millions still go hungry?**

4. **If diet shifts and waste reduction can close much of the calorie gap, why do you think these solutions are so difficult to implement?**

5. **What assumptions does your model make about technology development, and how sensitive are the results to those assumptions?**

# 8 STEM Challenge: Design a 2058 Global Food Production Plan

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## YOUR ENGINEERING MISSION

Design an agricultural strategy that feeds 10 billion people by 2058 while maintaining soil health, protecting biodiversity, and reducing agriculture's greenhouse gas emissions by 30%.

### THE SCENARIO

The United Nations Food and Agriculture Organization has convened your team of agricultural scientists to propose a 2058 global food security plan. World population will reach 10 billion, demand for food calories will increase by 50%, and climate change is reducing yields in many tropical regions. Your plan must increase food production without expanding farmland into remaining forests and without degrading soil below sustainable levels.

### Engineering Design Thinking

1. WHAT SPECIFIC AGRICULTURAL TECHNOLOGIES AND PRACTICES WILL YOUR PLAN DEPLOY, AND WHAT MODEL EVIDENCE SUPPORTS THEIR EFFECTIVENESS?

2. HOW WILL YOUR PLAN ADDRESS THE WATER SCARCITY CHALLENGE IN REGIONS WHERE AQUIFERS ARE BEING DEPLETED?

3. WHAT DIETARY SHIFTS DOES YOUR MODEL SUGGEST COULD REDUCE THE AMOUNT OF CROPLAND NEEDED FOR ANIMAL FEED?

4. HOW WILL YOUR PLAN BALANCE THE NEEDS OF SMALL-SCALE FARMERS IN DEVELOPING NATIONS WITH INDUSTRIAL AGRICULTURE IN DEVELOPED NATIONS?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Agricultural Scientists research ways to improve crop yields, soil health, and farming sustainability, working at universities, USDA, and agribusiness companies, earning $65,000–$120,000/year. Food Systems Analysts who model global food security for international organizations like the UN FAO earn $75,000–$140,000/year, designing policies that affect billions of people's access to nutrition.